can I run rerobot from github on my M5 macbook pro? How?

You can run (often misremembered or referred to as "rerobot") on an M-series Apple Silicon MacBook Pro, including for simulation, training, and data visualization. [1, 2]
However, running it for physical robot hardware control (like direct USB/serial arm teleoperation) depends on whether your specific robot drivers and motor SDKs support macOS. [1, 2]

How to Install and Run LeRobot on a Mac

1. **Prerequisites**
   - Ensure you have Git and **Python 3.10** (or a compatible version via Conda/Miniforge) installed.
   - Apple Silicon (M-series) Macs run PyTorch via the native mps (Metal Performance Shaders) backend for hardware acceleration. [1, 2]
2. **Clone the Repository**
   Open your Terminal and clone the official repository:bash

   git clone https://github.com/huggingface/lerobot.git
3. cd lerobot
4.

   Use code with caution.

5. **Create a Virtual Environment & Install Dependencies**
   Set up a Python environment and install the package:bash

   conda create -n lerobot python=3.10
6. conda activate lerobot
7. pip install -e .
8.

Use code with caution.

Note: Install any extra environment dependencies you need (such as simulation or specific vision extras) as noted in the LeRobot GitHub Documentation. [1]

9. **Run Simulations or Datasets**
   You can explore pre-trained models, inspect datasets, or run simulation environments directly on your MacBook without needing a physical robot attached. [1]

If you meant a different specific repository or are trying to connect a **particular robotic hardware piece** (like an SO-100 or Reachy arm) to your MacBook Pro, let me know which device you are using so I can check its macOS driver compatibility.

AI responses may include mistakes. Learn more

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GitHub

GitHub - huggingface/lerobot: 🤗 LeRobot: Making AI for Robotics more accessible with end-to-end learning · GitHub
Overview: Hugging Face LeRobot is an open-source, PyTorch-based Python library for end-to-end real-world robotics and physical AI.

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GitHub

GitHub - WeGo
LeRobot: State-of-the-art AI for real-world robotics 🤗 LeRobot aims to provide models, datasets, and tools for real-world robotic...

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  GitHub

  RainbowRobotics/rby1-lerobot - GitHub
  Rainbow Robotics RB-Y1 is a bimanual robot designed for physical AI research. This repository provides a LeRobot plugin for contro...

Show all

without a physical robot in place what can I do after I do the above installations? For example can I run one of the humanoid robots in simulation, either through lerobot or from elsewhere? can I RL-train it or train it otherwise for certain tasks and can I do all of it on my macbook pro M5? what is the maximum I can do?

Yes, you can do an immense amount of robotics development entirely in simulation on your M5 MacBook Pro without ever touching a physical robot. In fact, this is how the majority of modern AI robotics research is done before deploying to hardware.

The **M5 Pro chip's massive Unified Memory architecture** gives you a major advantage: you can fit enormous vision-based data models entirely into memory without hitting the VRAM walls common on Windows laptops.

The breakdown below details the absolute maximum you can achieve, what frameworks to use, and exactly where you will hit a wall due to hardware differences.

1. The Absolute Maximum You Can Do Natively

You can build, simulate, and train a complete humanoid or robotic system from your Mac by utilizing three specific pillars:

🤖 Imitation Learning & Dataset Evaluation (Via LeRobot)

Since you've already installed LeRobot, you can do state-of-the-art **Imitation Learning (Behavior Cloning)**.

- **What you can do:** You can download massive, real-world open-source datasets (like those from Hugging Face or the Aloha project). You can

write and run scripts to preprocess the dataset, visualize the camera streams, and train neural networks (like **Diffusion Policy** or **ACT**) entirely on your Mac's GPU using PyTorch's native mps (Metal Performance Shaders) backend.
- **The Sim:** LeRobot has native support for virtual simulation environments (like gym-pusht or lerobot-env). You can train an AI policy in these environments and watch the virtual robot perform the task on your screen.

🌌 **High-Parallel Reinforcement Learning (Via Genesis)**
While older robotics simulators rely entirely on NVIDIA CUDA, Genesis **is a major breakthrough framework built with native Apple Metal API support**.

- **What you can do:** You can import humanoid .urdf or .mjcf robot files (like Unitree H1 or Figure 01) directly into Genesis.
- **The Sim:** Genesis runs completely on your Mac GPU and features an ultra-fast, photo-realistic ray-tracer. You can spin up **hundreds of parallel humanoid robots** simultaneously on your M5 Pro, running complex rigid-body physics, and train them via Reinforcement Learning (RL) to walk, jump, or recover from falls.

🏋️ **Traditional RL (Via MuJoCo)**
DeepMind's MuJoCo physics engine compiles natively for macOS. You can run standard locomotion tasks (like the classic OpenAI Gym Humanoid-v4) to train reinforcement learning agents locally using your CPU and GPU.

2. The Core Technical Bottleneck: NVIDIA CUDA
The only place you will hit a wall on a MacBook Pro is the NVIDIA Omniverse **/** Isaac Sim **ecosystem**.

Standard NVIDIA Isaac Lab is the industry standard for reinforcement learning, but it is deeply hardcoded to require an NVIDIA RTX graphics card and CUDA drivers. It will not launch natively on macOS.

How to bypass this wall:
1. **The Community Route:** There are community forks like IsaacLab-mlx

which explicitly strip out the NVIDIA dependencies and port the math to Apple's native MLX framework so you can run basic environment loops on a Mac.
2. **The Cloud Pro Route (Industry Standard):** Most developers who prefer macOS use a hybrid setup. You code locally on your M5 MacBook Pro, push the script to a headless Linux cloud server (like AWS, Lambda Labs, or a home desktop with an RTX card), and use the NVIDIA Isaac Sim WebRTC Streaming Client on your Mac. This allows you to view the live, 3D streaming simulation window right on your Mac desktop while the heavy CUDA math processes remotely.

Summary Checklist for a Robot-Free Start

If you want to start seeing virtual robots move on your screen today without buying hardware, follow this progression:

1. **Test LeRobot's virtual training:** Run the native LeRobot training examples using a virtual environment (like pusht) to verify your PyTorch Metal backend is accelerating correctly.
2. **Install Genesis:** Go to the Genesis GitHub repository and install it. Run their humanoid or robotic arm example script. It will open a beautiful, ray-traced window showing a robot reacting to physics entirely powered by your M5 Pro.
3. **Download Datasets:** Use LeRobot to pull down public datasets from the internet to see how humanoids process visual and motor tokens.

Would you like to start with **Imitation Learning** (training a virtual robot by showing it data via LeRobot) or **Reinforcement Learning** (letting a robot learn via trial-and-error physics in Genesis)? I can provide the exact terminal commands to launch the virtual simulation for whichever path you prefer.